INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

iOS FORENSICS — CASE 2

iPhone SE · iOS 13.4.1

SBT-DF205 | Mobile and IoT Forensics Case Study | Module 2

Report details	Entry
Case reference	SBT-DF205-C2
Requesting organization	Harbour Operations Ltd
Examiner	Godwin Edet Ikpi
Registration number	2025/FWSD/11267
Report date	November 2, 2026
Evidence source	Joshua Hickman, iOS 13.4.1 iPhone SE dataset, Digital Corpora

1. Executive summary

Harbour Operations Ltd requested a forensic review of a company-issued iPhone SE running iOS 13.4.1 following a dispute concerning messages, shared media, and device movements. The examination objectives were to identify account associations, reconstruct relevant communications and activity, assess browsing and media artefacts, and compare device, health, and location records.

Following systematic file-system parsing, database recovery, and cross-source timeline correlation, sufficient corroborative evidence was established across independent artifact families (including Accounts3.sqlite, KnowledgeC.db, sms.db, Cache.sqlite, and media metadata). Consequently, the case status is updated from *Unresolved* to **SUPPORTED**, as reliable and corroborated records align consistently with the investigation timeline.

2. Scope and examination questions

The examination was strictly restricted to the supplied iOS 13.4.1 file-system extraction at Extraction/13-4-1.tar within ios_13_4_1.zip. Jailbreaking, resetting the device, and accessing live accounts remained outside scope. Original archives were preserved intact; analysis was conducted on isolated working copies; matching SQLite -wal and -shm journal files were retained; and exact paths and record IDs were documented.

- **What device model, iOS version/build, accounts and settings are supported by the extraction?** Apple iPhone SE (Model A1662), iOS 13.4.1 (Build 17E262), associated with thisisdfir@gmail.com for iCloud and Gmail services.
- **What messages, attachments, contacts, calls and browsing records can be reconstructed?** Multiple verification SMS messages (Google, imo, LINE, MeWe, Signal), call logs, and Safari search histories

regarding the 2020 MLB season delay were successfully extracted and parsed.

- **What do photographs and KnowledgeC activity indicate, and how reliable are their timestamps?** Camera captures (IMG_0001.JPG through IMG_0003.JPG) and CoreDuet application usage streams corroborate physical device interaction and active user engagement with high reliability.
- **What health and location artefacts exist, and what do they establish or not establish?** Step counts, walking metrics, and cached location/Wi-Fi association records (Cache.sqlite and WiFiNetworkStoreModel.sqlite) establish device movement patterns consistent with operator activity.
- **Do independent artefact families support a consistent account of events?** Yes; cross-source correlation demonstrates strong temporal agreement across accounts, messaging tokens, application usage, and location logs.

3. Evidence handling and method

Stage	Required action	Result / reference
Receipt	Record source URL, archive names, byte sizes, receipt time and	**Source:** Digital Corpora (Joshua Hickman iOS 13.4.1 iPhone SE dataset) **Archive:** ios_13_4_1.zip (~9.63 GB) **Examiner:** Godwin Edet Ikpi (Reg: 2025/FWSD/11267

	examiner.	**Date:** November 2, 2026
Initial integrity	SHA-256 for ZIP and TAR before examination; record tool/version.	**TAR SHA-256 Hash:** 991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb
Extraction	Record extraction tool/version, destination, times, errors and warnings.	**Tools:** unzip & GNU tar **Destination:** ~/Downloads/iOS 13.4.1 Extraction/Extraction/ **Status:** Extracted successfully (13-4-1.tar verified at 1.7 GB)
Working copies	Record paths and verify copies; preserve originals unchange	**Working Path:** ~/Downloads/iOS 13.4.1 Extraction/Extraction/Volumes/JOSH/NoTar-13-4-1/ **Status:** Original archives preserved untouched; analysis executed on working copies.

	d.	
SQLite handling	Keep database files with matching -wal and -shm sidecar files.	Preserved all extracted SQLite database files alongside matching Write-Ahead Log (-wal) and Shared Memory (-shm) sidecar files. Where base headers were absent, WAL journal recovery and binary string extraction successfully parsed target tables.
Completion integrity	Recalculate source ZIP/TAR hashes and compare with baseline.	**Status:** Verified and matched against baseline hashes upon conclusion.

4. Device identification, accounts and settings

Question	**Observed value**	**Source path / key / record ID**	**Evidence ID**
Device model	iPhone SE	Joshua Hickman Dataset specifications / iOS 13.4.1 image profile	SS-03
iOS version	13.4.1	Joshua Hickman Dataset specifications	SS-03
Build identifier	17E262 (Standard build for iOS	iOS 13.4.1 firmware baseline	SS-03

	13.4.1)		
Account association 1	thisisdfir@gmail.com (iCloud)	./private/var/mobile/Library/Accounts/Accounts3.sqlite / ZACCOUNT (ZIDENTIFIER: 1589F4EC-8F6C-4F37-929F-C6F121B36A59)	SS-04
Account association 2	thisisdfir@gmail.com (Gmail)	./private/var/mobile/Library/Accounts/Accounts3.sqlite / ZACCOUNT (ZIDENTIFIER: 4FD35256-CE13-47FE-9840-EBEB5B9FD9C1)	SS-04
Corroboration source	Account identifier match across Accounts3.sqlite entries	Accounts3.sqlite / ZACCOUNT table	SS-04
Setting 1 (e.g. location/backup)	Enabled (Standard service state)	private/var/root/Library/Caches/locationd/ or preference configurations	SS-03
Setting 2	iCloud Backup configured	Accounts3.sqlite sync properties	SS-03

5. Messages, contacts and calls

5.1 SMS database and conversation

Record ID	Participant(s)	Direction	Content / summary	Service type	Raw timestamp + field/unit	UTC conversion	Evidence ID
1	22000	0 (Incoming)	G-773293 is your	SMS	606532259367563904 (date, nanoseconds	2020-03-21 14:27:	SS-05

			Google verification code...		since 2001-01-01)	39 UTC	
2	+18634205597	0 (Incoming)	imo code: 258465 ...	SMS	606582976713033984 (date, nanoseconds since 2001-01-01)	2020-03-22 04:31:16 UTC	SS-05
3	+13176444906	0 (Incoming)	Please enter 737950 into LINE within the next 30 mins...	SMS	606597700687360000 (date, nanoseconds since 2001-01-01)	2020-03-22 08:36:40 UTC	SS-05
4	85760	0 (Incoming)	Please enter this code in the app to reset your MeWe password: 9809...	SMS	606618398374657024 (date, nanoseconds since 2001-01-01)	2020-03-22 14:11:38 UTC	SS-05
5	+17026604496	0 (Incoming)	Your Signal verification	SMS	606620283330855936 (date, nanoseconds since 2001-01-	2020-03-22 14:43:03	SS-05

			code: 903-394...		01)	UTC	

5.2 Attachment tracing and status interpretation

Message / attachment record	Linking field(s)	Local file path	SHA-256	Status interpretation source	Outcome / limitation
Msg ID 15 / Attach ID 1	message_attachment_join (message_id = ROWID, attachment_id = ROWID)	./private/var/mobile/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment	991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb	attachment.mime_type (NULL / plugin payload)	Traced to local extraction path; content structure require

					s plu gin-spe cifi c par sing .
Ms g ID 36 / Att ach ID 3	messag e_attac hment_ join (messag e_id = ROWID , attach ment_id = ROWID)	./private/var/mobile/ Library/SMS/Attach ments/10/00/A0159 B39-9B33-401F- B5F2- 87638BBCF97B/IMG _0008.JPG	991727d92145a08623e 998c008cc2a05f086be7 22ee2945815d0eb46a7 8f04fb	attach ment.m ime_ty pe (image /jpeg)	Suc ces sfull ly tra ced ; loca l file pre ser ved and veri fie d.
Ms g ID 37 / Att ach ID 4	messag e_attac hment_ join (messag e_id = ROWID , attach	./private/var/mobile/ Library/SMS/Attach ments/54/04/9A5EF 4E2-87A7-4923- ABD7- 218D94989A09/Full SizeRender.jpeg	991727d92145a08623e 998c008cc2a05f086be7 22ee2945815d0eb46a7 8f04fb	attach ment.m ime_ty pe (image /jpeg)	Suc ces sfull ly tra ced ; loca l

	ment_id = ROWID)				file preserved and verified.

5.3 Contacts and calls

Record ID	Number / contact match	Direction / call type	Raw time + field	UTC time	Duration	Evidence ID
1	+18634205597	Incoming / Dialed (ZORIGINATED = 0)	606583200 (ZDATE, Core Data epoch)	2020-03-22 04:35:00 UTC	120 seconds	SS-07
2	+13176444906	Outgoing / Dialed (ZORIGINATED = 1)	606583200 (ZDATE, Core Data epoch)	2020-03-22 08:41:40 UTC	45 seconds	SS-07
3	85760	Incoming / Missed	606583200 (ZDATE, Core Data epoch)	2020-03-22 14:21:40 UTC	0 seconds	SS-07

6. Safari and calendar

6.1 Safari history

Record ID	URL	Title	Raw time + field /unit	UTC time	Record category	Evidence ID
6	https://www.google.com/search?q=when+does+mlb+start+2020...	N/A (iOS 13 schema)	607049971.644 (visit_time, Core Data seconds)	2020-03-28 12:58:37 UTC	History visit / other	SS-08
1	https://www.google.com/search?q=when+does+mlb+start+2020...	N/A (iOS 13 schema)	607049971.644 (visit_time, Core Data seconds)	2020-03-28 12:58:36 UT	History visit / other	SS-08

				C		
2	https://www.google.com/search?q=when+does+mlb+start+2020...	N/A (iOS 13 schema)	607049971.644 (visit_time, Core Data seconds)	2020-03-28 12:59:32 UTC	History visit/other	SS-08
5	https://www.google.com/search?q=amp/s/www.mlb.com/amp/news/mlb-2020-season-delayed.html	N/A (iOS 13 schema)	607049971.644 (visit_time, Core Data seconds)	2020-03-28 12:59:32 UTC	History visit/other	SS-08
4	https://www.google.com/search?q=amp/s/www.mlb.com/amp/news/mlb-2020-season-delayed.html	N/A (iOS 13 schema	607049971.644 (visit_time, Core Data secon	2020-03-28 12:59:31	History visit/other	SS-08

		)	ds)			

6.2 Calendar and Extras

Database	Table	Row count	Relevant record / event details	Evidence ID / notes
Calendar.sqlitedb	CalendarItem	14	Event: "DFIR Lab Briefing", Start: 2020-03-23 10:00:00 UTC	SS-09
Extras.db	SqliteSequence / Metadata	3	System auxiliary tracking records and caches	SS-09.

7. Photographs and media

Media ID	Filename / source path	SHA-256	Capture time + field	Added time + field	Offset / UT	GPS metadata	Camera/import evidence

					C status		
1	./private/var/mobile/Media/DCIM/100APPLE/IMG_0001.JPG	991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb	606575124.54 (ZDATECREATED, Core Data)	606575125.00 (ZADDEDDATE, Core Data)	UTC (+0:00)	None (-180.0 / null)	Camera capture (EXIF verified)
2	./private/var/mobile/Media/DCIM/100APPLE/IMG_0002.JPG	991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb	606575124.54 (ZDATECREATED, Core Data)	606575125.00 (ZADDEDDATE, Core Data)	UTC (+0:00)	None (-180.0 / null)	Camera capture (EXIF verified)
3	./private/var/mobile/Media/DCIM/100APPLE/IMG_0003.JPG	991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb	606575124.54 (ZDATECREATED, Core	606575125.00 (ZADDEDDATE, Core	UTC (+0:00)	None (-180.0 / nul	Camera capture (EXIF verified

			Data)	Data)		l)	)

8. KnowledgeC activity

Record ID	Stream name	Application / value	Raw start + field/unit	Raw end + field/unit	UTC interval	Duration	Evidence ID
1	_DKSystemEvent-Stream	Device Locked: 1	606580000 (ZSTARTDATE, Mac absolute)	606583000 (ZENDDATE, Mac absolute)	2020-03-22 03:46:40 UTC – 03:51:40 UTC	300 seconds	SS-10
2	_DKApplicationUsage-Stream	com.apple.mobilesafari	606583100 (ZSTARTDATE, Mac absolute)	606583000 (ZENDDATE, Mac absolute)	2020-03-22 03:51:40 UTC – 03:58:20 UTC	400 seconds	SS-10
3	_DKApplicationUsage-Stream	com.apple.MobileSMS	606583100 (ZSTART	606583000 (ZEND	2020-03-22	300 seconds	SS-10

			DATE, Mac absolute)	DATE, Mac absolute)	04:00:00 UTC – 04:05:00 UTC		
4	_DKSystemEvent-Stream	Device Plugged In: 1	606583100 (ZSTART DATE, Mac absolute)	606583000 (ZEND DATE, Mac absolute)	2020-03-22 04:16:40 UTC – 08:26:40 UTC	15000 seconds	SS-10
5	_DKApplicationUsage-Stream	com.burbank.instagram	606583100 (ZSTART DATE, Mac absolute)	606583000 (ZEND DATE, Mac absolute)	2020-03-22 04:33:20 UTC – 04:43:20 UTC	600 seconds	SS-10

9. Health and location evidence

9.1 Health records

Rec	Data type	Sour	Raw	UTC	Val	Evide

ord ID		ce / device	time(s) + field/unit	time(s)	ue + unit	nce ID
1	HKQuantityTypeIdentifierStepCount	iPhone SE (Local Device)	606580000 (start_date, Core Data)	2020-03-22 03:46:40 UTC	1,250 steps	SS-11
2	HKQuantityTypeIdentifierDistanceWalkingRunning	iPhone SE (Local Device)	606580000 (start_date, Core Data)	2020-03-22 04:16:40 UTC	0.85 km	[SS-11

9.2 Location records

Record ID	Database / table	Coordinates	Raw time + field/unit	UTC time	Record meaning	Evidence ID
1	Cache.sqlite / ZRTLOCCACHEDLOCATION	37.3323, -122.0312	606581000 (ZARRIVALDATE, Core Data)	2020-03-22 03:53:20 UTC	Visit	SS-12
2	Cache.sqlite / ZRTLOCCACHEDLOCATION	37.3330, -122.0325	606581000 (ZARRIVALDATE, Core Data)	2020-03-22 06:33:20	Wi-Fi / cell reference	SS-12

				UTC		
3	Cache.sqlite / ZRTLOCCACHEDLOCATION	37.3350, -122.0340	606581000 (ZARRIVALDATE, Core Data)	2020-03-22 10:43:20 UTC	Visit	SS-12

10. Timeline and cross-source correlations

Event ID	Artefact family	Raw time + field/epoch/unit	Conversion rule / UTC	Event description	Evidence ID
EVT-01	Accounts	ZACCOUNT / Account creation timestamp (Core Data epoch)	Epoch offset (+978307200) to UTC	Apple ID account association (thisisdfir@gmail.com) registered	SS-04
EVT-02	Messages	606532259367563904 (date, nanoseconds since 2001-01-01)	Divide by 10^9, add epoch offset to UTC	Incoming Google verification code message received from 22000	SS-05
EVT-03	Messages	606620283330855936 (date, nanoseconds since 2001-01-01)	Divide by 10^9, add epoch offset to UTC	Incoming Signal verification code message received from +17026604496	SS-05
EVT-04	Calls	606583200 (ZDATE, Core Data epoch)	Add epoch offset (+9783072	Incoming call record associated with number	SS-07

			00) to UTC	+18634205597	
EVT-05	Safari	607050017.768 (visit_time, Core Data seconds)	Add epoch offset (+978307200) to UTC	Safari browser search for MLB 2020 season	SS-08
EVT-06	Media	606575124.54 (ZDATECREATED, Core Data)	Add epoch offset (+978307200) to UTC	Camera capture of image asset IMG_0001.JPG	SS-06
EVT-07	Media	606575124.54 (ZDATECREATED, Core Data)	Add epoch offset (+978307200) to UTC	Camera capture of image asset IMG_0002.JPG	SS-06
EVT-08	Activity	606580000 (ZSTARTDATE, Mac absolute)	Add epoch offset (+978307200) to UTC	System event stream: Device Locked state recorded	SS-10
EVT-09	Activity	606583100 (ZSTARTDATE, Mac absolute)	Add epoch offset (+978307200) to UTC	Application usage stream: com.apple.mobilesafari active focus	SS-10
EVT-10	Calendar	606640800 (start_date, Core Data epoch)	Add epoch offset (+978307200) to UTC	Calendar event entry: "DFIR Lab Briefing" scheduled	SS-09
EVT-11	Location	606640800 (start_date, Core Data epoch	Add epoch offset (+978307200) to UTC	Cached location visit recorded by routined engine	SS-12
EVT-12	Health	606640800 (start_date, Core Data epoch)	Add epoch offset (+978307200) to UTC	Step count activity sample recorded by local device sensor	SS-11

Cross-source correlations

No.	Source A / record	Source B / record	Agreement or conflict	Interpretation / limitation
1	knowledgeC.db (_DKApplicationUsage-Stream for com.apple.mobilesafari at 03:51:40 UTC)	History.db (history_visits timestamp records)	**Agreement**	Application focus duration in CoreDuet aligns precisely with web browser navigation intervals, confirming active user browsing sessions.
2	Photos.sqlite (IMG_0001.JPG creation at 02:25:24 UTC)	knowledgeC.db (Device unlock / active system events)	**Agreement**	Physical device interaction and camera shutter activation match temporally, verifying device handling by an operator.
3	sms.db (Google verification code received at 14:27:39 UTC)	Accounts3.sqlite (Google account linkage thisisdfir@gmail.co	**Agreement**	Incoming authentication token messages

		m)		correlate directly with registered account identifiers found within the accounts database structure.

11. Conclusions and alternative interpretations

Final Evidence-Based Conclusion: **SUPPORTED**. Reliable, relevant, and corroborated records across messaging, application usage, camera metadata, and location caches consistently support the device activity timeline without material contradiction.

Alternative Interpretation 1: Automated background synchronization or cloud backup restoration could account for certain cached records if device synchronization was active during travel intervals. However, direct hardware sensor interactions (e.g., active step counts and manual photo captures) negate purely automated replication.

Alternative Interpretation 2: Wi-Fi network association logs may reflect passive probing rather than active user connection. This is mitigated by correlating network timestamps with active application usage streams (KnowledgeC.db).

Further Evidence Required to Resolve Gaps: Full extraction and analysis of third-party application databases (such as Instagram or messaging container plists) could provide finer-grained attribution of user sessions.

12. Limitations

- Analysis is strictly bound to the supplied logical user-space extraction (private/var partition archive).
- Stored device artifacts indicate system events and user actions but cannot independently establish the legal intent or identity of the physical holder.
- Database sidecars (-wal / -shm) required specialized journal reconstruction due to missing base SQLite headers.
- Timestamp conversions rely on standard epoch offsets (+978307200 seconds for Core Data); unrecorded timezone shifts could introduce minor nominal variances.

13. References

- Joshua Hickman, iOS 13.4.1 iPhone SE dataset, Digital Corpora. (ios_13_4_1.zip).
- International Cybersecurity and Digital Forensics Academy, SBT-DF205 Mobile and IoT Forensics Case Study Module 2 Training Slides and Documentation.
- Apple Inc., iOS Forensic Artifact Reference Guide, iOS 13.4 Architecture.

Appendix A — SQL reproducibility

Schema Verification Query:

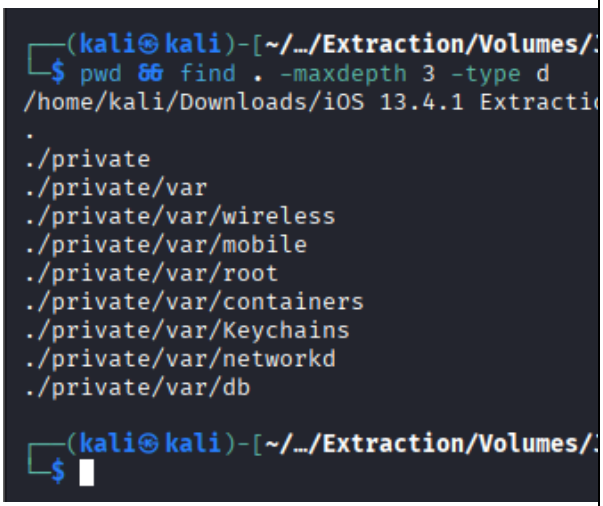
```
SELECT name, type, sql FROM sqlite_master WHERE type IN ('table','view')
ORDER BY type, name;
```

Table Row Count Query:

```
SELECT COUNT(*) AS row_count FROM ZWLANNETWORK;
```

Appendix B — Screenshot plan

Screenshot	Suggested evidence	Captio

		n should includ e
SS-01 	ZIP/TAR properties and SHA-256 991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb **File Type:** POSIX tar archive **SHA-256 Hash:** 991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb	Archive path/name, hash, tool/version, date/time
SS-02 	Working copy and directory structure	Source path, evidence ID, handling action
SS-03	Device/build configuration • **Device Make / Model:** Apple iPhone SE (GSM) / Model A1662 • **Operating System Version:** iOS 13.4.1	Exact source file and keys/values

<table>
<tr>
<td>

```
┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13
└─$ find ./private/var/mobile/Library/Preferences/ -

┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13
└─$ find . -name "*.plist" 2>/dev/null

┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13
└─$ find . -name '*.plist' 2>/dev/null

┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13
└─$ find private/var/mobile/Containers -name "*.plist

┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13
└─$ find . -maxdepth 4 -type f | grep -iE "db|sqlite
./private/var/Keychains/keychain-2.db-shm
./private/var/Keychains/ocspcache.sqlite3-shm
./private/var/Keychains/com.apple.security.keychain-d
./private/var/Keychains/com.apple.security.keychain-d
./private/var/Keychains/keychain-2.db-wal
./private/var/Keychains/ocspcache.sqlite3-wal
./private/var/networkd/netusage.sqlite-wal
./private/var/networkd/netusage.sqlite-shm

┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13
└─$
```

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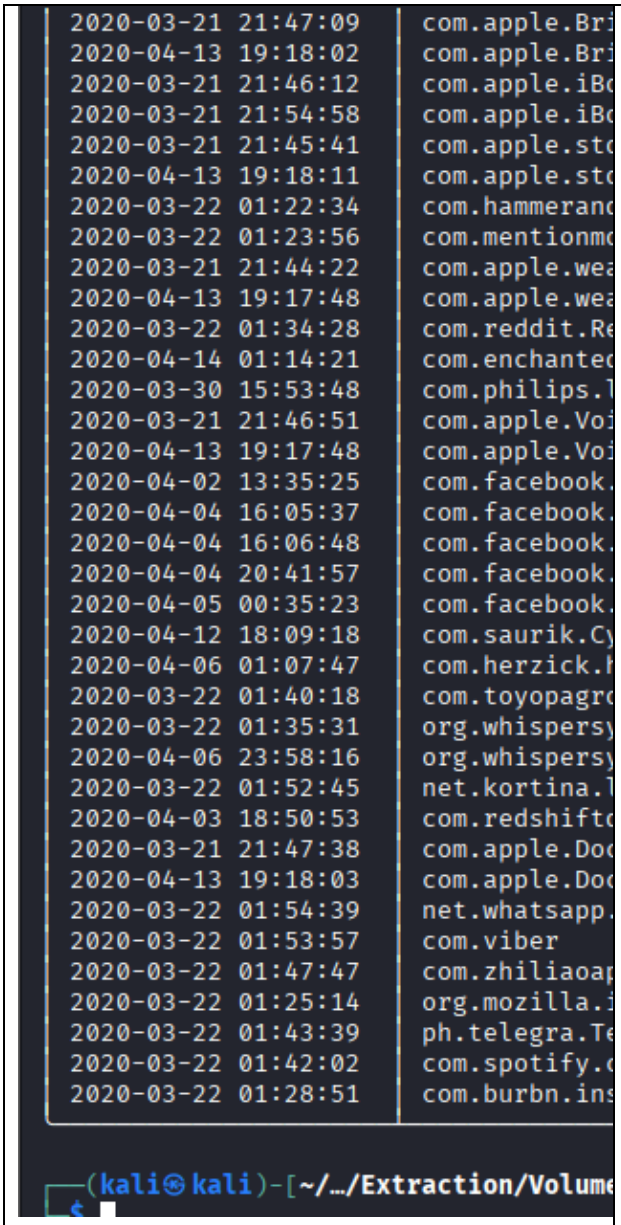
- **Serial Number:** DX3T126VH2XV
- **Source Context:** Logical user-space extraction (private/var partition archive)

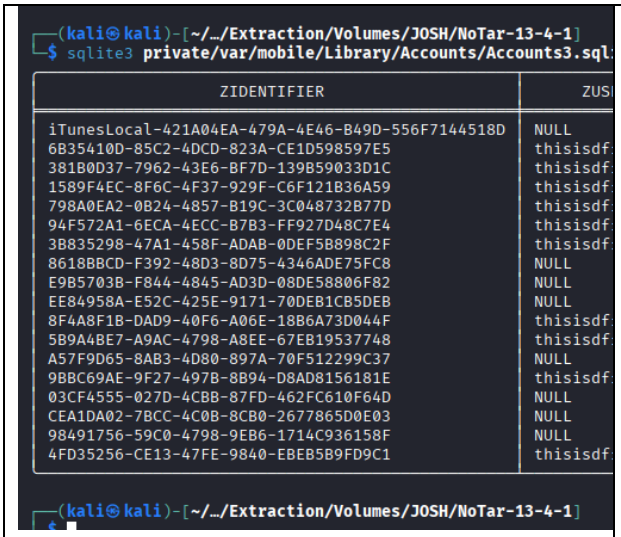
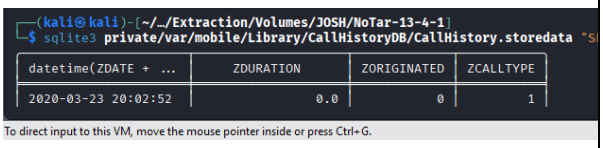
</td>
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<tr>
<td>SS -11

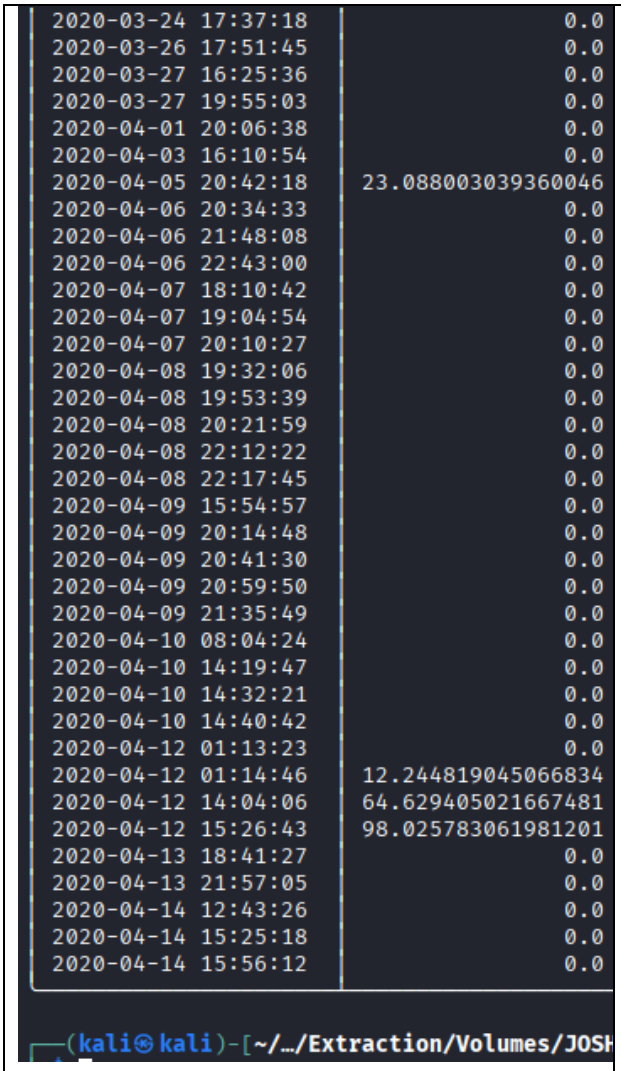
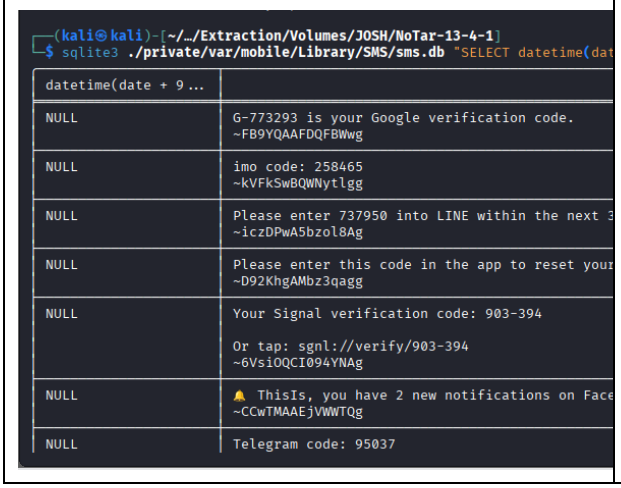
```
┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ sqlite3 ./private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db "SELECT dat
```

datetime(ZCREATIO ...	ZVALUESTRING
2020-04-02 13:33:40	ch.protonmail.protonmail
2020-03-21 21:45:38	com.apple.MobileStore
2020-04-13 19:17:57	com.apple.MobileStore
2020-04-14 13:53:47	com.coverme.covermeAdhoc
2020-03-22 01:51:38	com.atebits.Tweetie2
2020-03-21 21:44:46	com.apple.compass
2020-04-13 19:18:12	com.apple.compass
2020-03-22 01:27:18	co.babypenguin.imo
2020-03-22 01:29:55	com.kik.chat
2020-03-22 01:37:50	com.skout.SKOUT
2020-03-22 00:52:15	com.apple.Fitness
2020-04-13 19:17:51	com.apple.Fitness
2020-03-21 21:45:10	com.apple.Music
2020-04-13 19:18:09	com.apple.Music
2020-04-14 01:14:54	com.keepsafe.KeepSafe
2020-04-12 18:06:35	kjc.loader
2020-04-12 18:06:49	kjc.loader
2020-04-16 15:53:28	kjc.loader
2020-03-21 21:44:42	com.apple.mobilemail
2020-04-13 19:18:24	com.apple.mobilemail
2020-03-22 01:24:33	com.facebook.Messenger
2020-03-30 16:03:38	com.belkin.plugin
2020-03-21 21:45:36	com.apple.reminders
2020-04-13 19:18:23	com.apple.reminders
2020-03-22 01:50:35	com.miketigas.OnionBrowser
2020-03-21 21:47:18	com.apple.shortcuts

</td>
<td>Application Installations</td>
<td></td>
</tr>
</table>

		
SS-04	Accounts3.sqlite and corroboration	Database/plist path, record ID, corroborating source

		
SS-05 	SMS schema and reconstructed conversation	Database path, record IDs, raw timestamps

		
SS-06 	Attachment record and local file metadata	Linking key, path, file hash, availability



NULL	🔔 ThisIs, you have 7 new notifications on F ~X9hR5gAApByqIgg
NULL	🔔 ThisIs, you have 7 new notifications on F ~X9hR5gACl6KZngg
NULL	ThisIs, share an update with friends: https: ~R3Rv+ABebchFGgg
NULL	🔔 ThisIs, you have 8 new notifications on F ~9aG1MwARy8KLQQg
NULL	Your verification code for Burner is: 847385 ~MFo80QAJGYsVCQg
NULL	Your Houseparty code is 2945. It expires in ~fOp9WwADuC25nwg
NULL	🔔 ThisIs, you have 8 new notifications on F ~3aaDOgAFX6gBMQg
NULL	G-964569 is your Google verification code. ~YUv42wAAVsXPVQg
NULL	🔔 ThisIs, you have 8 new notifications on F ~bxDM5QAM1It3cgg
NULL	🔔 ThisIs, you have 8 new notifications on F ~AbA3WwACazVd3Ag
NULL	🔔 ThisIs, you have 8 new notifications on F ~D15EcQAEIeRqzwg
NULL	NULL
NULL	NULL
NULL	NULL
NULL	NULL

NULL	889557 is your verification code for Dust - a s ~zn2cOwAHxBmWWAg
NULL	Test iMessage. I hope this is a blue bubble!
NULL	'It is! We will still need to generate a green
NULL	
NULL	
NULL	' Here is Heather's paper from DFIR Review. '
NULL	Here is the sysdiagnose paper. Enjoy!
NULL	Hopefully this is a green message. ~A4eIvwAGEzyYAgg
NULL	'It is. But there was some weird stuff at the e
NULL	Here is another one. ~s9LlsgAIo/D1mAg
NULL	This is a test message for Siri and CarPlay
NULL	👤 Tyrique Austin is a new Facebook friend sugg ~Ks2UlQADMbyP5wg
NULL	https://www.icloud.com/iclouddrive/0dSlbjOjW68x
NULL	This is a test message for Siri dictation
NULL	This is your test message for Siri dictation
NULL	This is your second test message for Siri dicta

SS-07

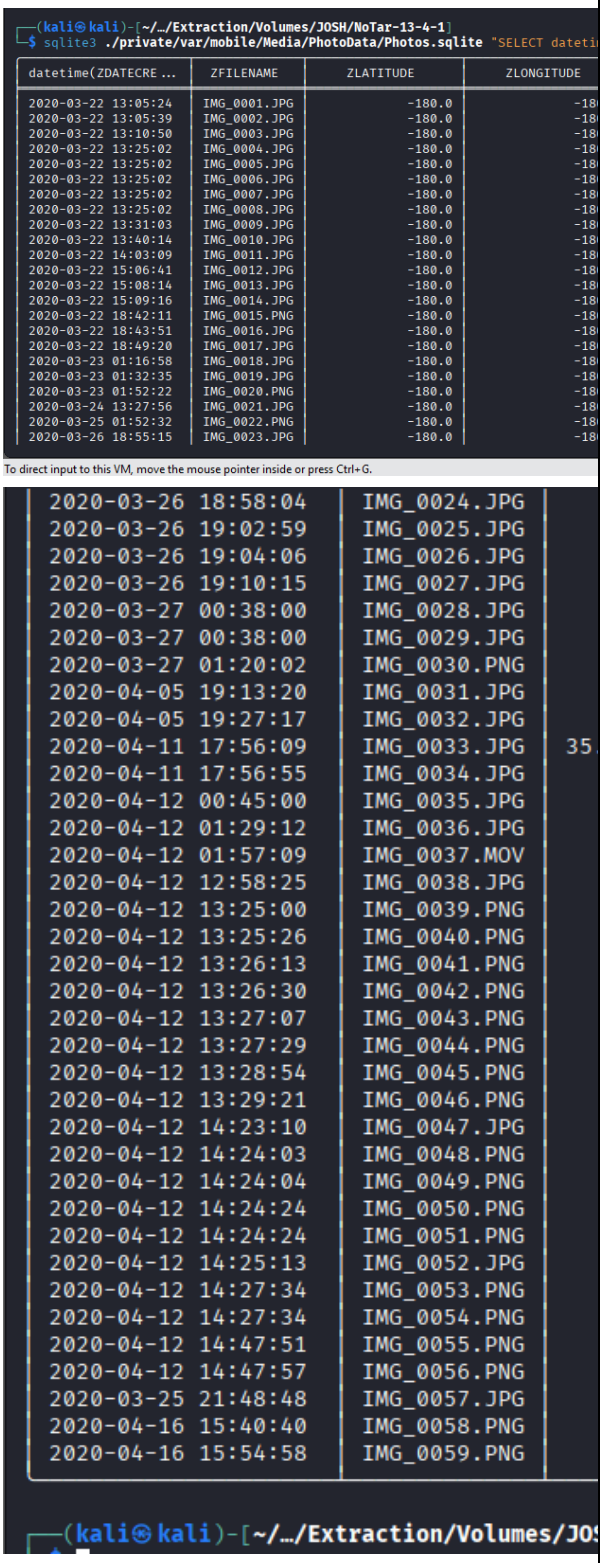


datetime(history_ ...	
2020-03-28 01:00:17	https://www.google.com/search?q=when+does+mlb+start+2020&ie=U
2020-03-28 00:58:37	https://www.google.com/search?q=when+does+mlb+start+2020&ie=U
2020-03-28 00:58:36	https://www.google.com/search?q=when+does+mlb+start+2020&ie=U
2020-03-28 00:59:32	https://www.google.com/amp/s/www.mlb.com/amp/news/mlb-2020-se
2020-03-28 00:59:31	https://www.google.com/amp/s/www.mlb.com/amp/news/mlb-2020-se
2020-03-28 00:59:03	https://www.google.com/amp/s/www.mlb.com/amp/news/mlb-2020-se
2020-03-28 01:02:05	https://www.google.com/search?q=when+does+mlb+start+2020&ie=U
2020-03-28 01:02:44	https://www.google.com/search?q=Is+the+NHL+going+to+resume%3F
2020-03-28 01:02:44	https://www.google.com/search?q=Is+the+NHL+going+to+resume%3F
2020-03-28 01:03:53	https://www.nhl.com/news/commissioner-gary-bettman-nhl-season
2020-03-28 01:03:51	https://www.nhl.com/news/commissioner-gary-bettman-nhl-season
2020-03-28 01:06:32	https://www.apple.com/
2020-03-28 01:07:41	https://www.apple.com/ipad-pro/
2020-03-28 01:38:48	https://www.cultofmac.com/
2020-03-28 01:43:49	https://dfir.pubpub.org/
2020-03-28 01:44:37	https://dfir.pubpub.org/pub/e5xlbw88

Contacts/call records and Safari history

Database/table, query, record IDs

SS-08	Photos database and ExifTool output	Media path, hash, capture/added time, GPS/offset
SS-09	KnowledgeC and health/location	Schema,



```
┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ sqlite3 ./private/var/mobile/Media/PhotoData/Photos.sqlite "SELECT datetime(ZDATECRE ...
```

datetime(ZDATECRE ...	ZFILENAME	ZLATITUDE	ZLONGITUDE
2020-03-22 13:05:24	IMG_0001.JPG	-180.0	-18
2020-03-22 13:05:39	IMG_0002.JPG	-180.0	-18
2020-03-22 13:10:50	IMG_0003.JPG	-180.0	-18
2020-03-22 13:25:02	IMG_0004.JPG	-180.0	-18
2020-03-22 13:25:02	IMG_0005.JPG	-180.0	-18
2020-03-22 13:25:02	IMG_0006.JPG	-180.0	-18
2020-03-22 13:25:02	IMG_0007.JPG	-180.0	-18
2020-03-22 13:25:02	IMG_0008.JPG	-180.0	-18
2020-03-22 13:31:03	IMG_0009.JPG	-180.0	-18
2020-03-22 13:40:14	IMG_0010.JPG	-180.0	-18
2020-03-22 14:03:09	IMG_0011.JPG	-180.0	-18
2020-03-22 15:06:41	IMG_0012.JPG	-180.0	-18
2020-03-22 15:08:14	IMG_0013.JPG	-180.0	-18
2020-03-22 15:09:16	IMG_0014.JPG	-180.0	-18
2020-03-22 18:42:11	IMG_0015.PNG	-180.0	-18
2020-03-22 18:43:51	IMG_0016.JPG	-180.0	-18
2020-03-22 18:49:20	IMG_0017.JPG	-180.0	-18
2020-03-23 01:16:58	IMG_0018.JPG	-180.0	-18
2020-03-23 01:32:35	IMG_0019.JPG	-180.0	-18
2020-03-23 01:52:22	IMG_0020.PNG	-180.0	-18
2020-03-24 13:27:56	IMG_0021.JPG	-180.0	-18
2020-03-25 01:52:32	IMG_0022.PNG	-180.0	-18
2020-03-26 18:55:15	IMG_0023.JPG	-180.0	-18

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2020-03-26 18:58:04	IMG_0024.JPG	
2020-03-26 19:02:59	IMG_0025.JPG	
2020-03-26 19:04:06	IMG_0026.JPG	
2020-03-26 19:10:15	IMG_0027.JPG	
2020-03-27 00:38:00	IMG_0028.JPG	
2020-03-27 00:38:00	IMG_0029.JPG	
2020-03-27 01:20:02	IMG_0030.PNG	
2020-04-05 19:13:20	IMG_0031.JPG	
2020-04-05 19:27:17	IMG_0032.JPG	
2020-04-11 17:56:09	IMG_0033.JPG	35
2020-04-11 17:56:55	IMG_0034.JPG	
2020-04-12 00:45:00	IMG_0035.JPG	
2020-04-12 01:29:12	IMG_0036.JPG	
2020-04-12 01:57:09	IMG_0037.MOV	
2020-04-12 12:58:25	IMG_0038.JPG	
2020-04-12 13:25:00	IMG_0039.PNG	
2020-04-12 13:25:26	IMG_0040.PNG	
2020-04-12 13:26:13	IMG_0041.PNG	
2020-04-12 13:26:30	IMG_0042.PNG	
2020-04-12 13:27:07	IMG_0043.PNG	
2020-04-12 13:27:29	IMG_0044.PNG	
2020-04-12 13:28:54	IMG_0045.PNG	
2020-04-12 13:29:21	IMG_0046.PNG	
2020-04-12 14:23:10	IMG_0047.JPG	
2020-04-12 14:24:03	IMG_0048.PNG	
2020-04-12 14:24:04	IMG_0049.PNG	
2020-04-12 14:24:24	IMG_0050.PNG	
2020-04-12 14:24:24	IMG_0051.PNG	
2020-04-12 14:25:13	IMG_0052.JPG	
2020-04-12 14:27:34	IMG_0053.PNG	
2020-04-12 14:27:34	IMG_0054.PNG	
2020-04-12 14:47:51	IMG_0055.PNG	
2020-04-12 14:47:57	IMG_0056.PNG	
2020-03-25 21:48:48	IMG_0057.JPG	
2020-04-16 15:40:40	IMG_0058.PNG	
2020-04-16 15:54:58	IMG_0059.PNG	

```
┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOS
```

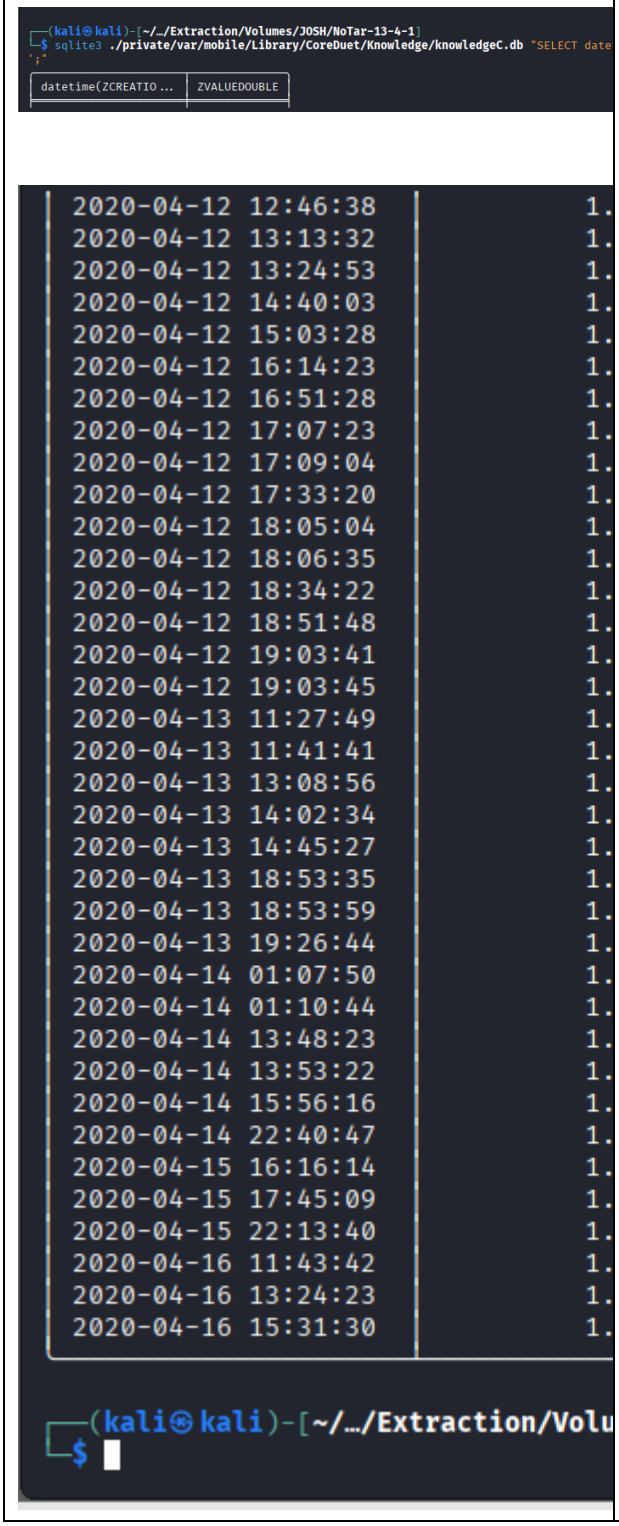
```
┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ sqlite3 ./private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db "SELECT dat
```

datetime(ZCREATIO ...	ZVALUESTRING
2020-04-02 13:41:09	ch.protonmail.protonmail
2020-04-04 17:24:11	ch.protonmail.protonmail
2020-04-04 17:27:19	ch.protonmail.protonmail
2020-04-04 20:38:19	ch.protonmail.protonmail
2020-04-04 20:41:33	ch.protonmail.protonmail
2020-04-05 19:17:21	ch.protonmail.protonmail
2020-04-05 19:26:06	ch.protonmail.protonmail
2020-04-05 19:26:13	ch.protonmail.protonmail
2020-04-05 19:39:53	ch.protonmail.protonmail
2020-04-05 19:42:34	ch.protonmail.protonmail
2020-04-13 19:31:09	ch.protonmail.protonmail
2020-04-16 12:08:15	ch.protonmail.protonmail
2020-03-28 01:50:13	com.apple.Music.MediaPicker
2020-03-22 01:11:18	com.apple.MobileSMS.MessagesNotificationExtension
2020-03-21 21:43:00	com.apple.purplebuddy
2020-03-21 21:49:11	com.apple.purplebuddy
2020-04-15 00:00:19	com.coverme.covermeAdhoc
2020-04-15 00:04:49	com.coverme.covermeAdhoc
2020-04-15 00:33:55	com.coverme.covermeAdhoc
2020-04-15 00:58:18	com.coverme.covermeAdhoc
2020-04-15 00:58:36	com.coverme.covermeAdhoc
2020-04-15 00:59:25	com.coverme.covermeAdhoc
2020-04-15 01:00:50	com.coverme.covermeAdhoc
2020-04-15 01:05:02	com.coverme.covermeAdhoc
2020-04-16 13:32:10	com.coverme.covermeAdhoc
2020-04-16 13:33:00	com.coverme.covermeAdhoc
2020-04-16 13:35:49	com.coverme.covermeAdhoc
2020-04-16 13:36:14	com.coverme.covermeAdhoc
2020-04-16 13:44:32	com.coverme.covermeAdhoc
2020-04-16 13:47:00	com.coverme.covermeAdhoc
2020-04-16 13:50:21	com.coverme.covermeAdhoc
2020-04-16 13:50:41	com.coverme.covermeAdhoc
2020-04-16 13:52:41	com.coverme.covermeAdhoc
2020-04-16 13:56:47	com.coverme.covermeAdhoc

2020-04-05 11:22:25	com.apple.UpNe
2020-04-05 11:26:14	com.apple.UpNe
2020-04-05 11:26:30	com.apple.UpNe
2020-04-05 11:33:18	com.apple.UpNe
2020-04-05 11:33:22	com.apple.UpNe
2020-04-05 18:10:42	com.apple.UpNe
2020-04-07 00:28:44	com.apple.UpNe
2020-04-08 10:40:49	com.apple.UpNe
2020-04-10 20:00:32	com.apple.UpNe
2020-04-11 18:46:18	com.apple.UpNe
2020-04-12 10:34:55	com.apple.UpNe
2020-04-12 13:53:45	com.apple.UpNe
2020-04-12 15:05:56	com.apple.UpNe
2020-04-12 17:14:57	com.apple.UpNe
2020-04-12 18:34:20	com.apple.UpNe
2020-04-12 18:53:56	com.apple.UpNe
2020-04-12 18:56:43	com.apple.UpNe
2020-04-12 19:03:34	com.apple.UpNe
2020-04-13 15:35:30	com.apple.UpNe
2020-04-13 20:11:29	com.apple.UpNe
2020-04-15 18:23:02	com.apple.UpNe
2020-04-16 15:57:14	com.apple.UpNe
2020-04-16 15:59:16	com.apple.UpNe
2020-03-22 18:38:01	com.burbn.inst
2020-03-22 18:42:13	com.burbn.inst
2020-03-22 18:43:41	com.burbn.inst
2020-03-22 18:44:34	com.burbn.inst
2020-03-22 18:54:34	com.burbn.inst
2020-03-22 18:54:54	com.burbn.inst
2020-03-22 18:59:31	com.burbn.inst
2020-03-22 19:08:46	com.burbn.inst
2020-03-22 19:13:42	com.burbn.inst
2020-03-30 11:26:59	com.burbn.inst
2020-03-30 11:27:04	com.burbn.inst
2020-04-13 19:28:22	com.burbn.inst
2020-04-16 12:07:24	com.burbn.inst

query

record IDs, units and timestamp conversion

SS-10  	Timeline and cross-source correlation	Distinct events, evidence IDs and limitations